RE22 – BioEnergy

The extensive experiment system brought to you by the Armfield BioEnergy kit enables you to reconstruct and understand the whole biomass cycle without any additional equipment.

Thus allows the exploration of the energetic use of biomass.

PRODUCT CODE: **RE22**

Categories: Educational, RE Series - Renewable Energy, RE22 Bio Energy

Tags: Anaerobic Degradation , BioEnergy Kit , Biomass , Biomass Cycle , Gas Collecting , Germination , Hydroculture , Hydrogen , Methane , RE22 , Renewable Energy

DESCRIPTION

The extensive experiment system brought to you by the Armfield BioEnergy kit enables you to reconstruct and understand the whole biomass cycle without any additional equipment.

A cultivation box and hydroculture allow the observation of the sprouting and growth of plants. Thereby the water and nutrient consumption can be analysed in the different growth phases.

Different experiments then show the aerobic as well as the anaerobic degradation of the biomass in a compost or biogas processes.

Thus allows the exploration of the energetic use of biomass.

TECHNICAL SPECIFICATIONS

- 1 x composter
- 1 x Gas collecting container
- 1 x Burner
- 1 x Tripod plant lighting
- 1 x Sprout box
- 1 x Box 6 L
- 1 x Tripod
- 1 x Seed set
- 1 x Rubber plug with tube

- 1 x ID tags
- 1 x Aluminium case 1710 silver
- 1 x Aluminium case 1710 blue
- 2 x Pot holder BioEnergy
- 1 x Propeller
- 1,5 x Silicone tube 4mm
- 1 x Switch 4mm
- 2 x Plant light
- 1 x Hose clamp
- 1 x Test lead black 25 cm
- 1 x Test lead red 25 cm
- 2 x Short-circuit plug
- 1 x Erlenmeyer flask 1000 ml
- 2 x Timer
- 1 x Air pump
- 2 Aeration stone
- 50 Net cup planter
- 1 EC meter
- 1 x Temperature logger
- 1 x Weight
- 1 x Tweezers
- 24 x Stopper red
- 1 x Nebulizer
- 2 x Plastic box Gratnells 75 mm deep
- 1 x Insert BioEnergy Rtg 1710
- 2 x Info sheet initial startup

FEATURES & BENEFITS

- Explore of the energetic use of biomass
- Cultivation box and hydro-culture
- Aerobic as well as the anaerobic degradation of the biomass
- Understand the whole biomass cycle
- Observe of the sprouting and growth of plants
- Analyse nutrient consumption in different growth phases

Experimental Content

BioEnergy experiments

- Germination of plant seeds
- Plant growth in a hydroculture
- Consumption of water and nutrients
- Aerobic degradation of biomass in a compost
- Anaerobic degradation of biomass to form hydrogen
- Anaerobic degradation of biomass to form methane

Related Curriculums

- Renewable Energies
- Chemical Engineering
- Environmental Engineering

Other products in the series

RE10: Advanced Photovoltaic Energy
RE12: Advanced Wind Energy
RE14: Advanced Fuel Cell Technology
RE16: Advanced Thermal Energy
RE20: Bio-Fuel
RE24: Advanced Battery Technology

Operational Conditions

- Storage Temperature: -10°C to +70°C
- Operating temperature range: +10°C to +50°C
- Operating relative humidity range: 0 to 95%, non-condensing

Requirements

Electrical supply: 110-230V AC 50-60Hz

- Level and stable work surface
- 1 x Bunsen burner (Not supplied by Armfield)

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

- **Volume:** 0.038m³
- **Gross weigh:** 14.6Kg

Dimensions

TRAY

- **Length:** 0.640m
- **Width:** 0.165m
- **Height:** 0.370m

Order Codes

RE22: BioEnergy

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